Academically Defendable Workflow

LR-FHSS Multi-Beam LEO Integration

- Focus: system model, channel model, mathematical flow
- Excludes: software command usage

Problem and Outputs

- Goal: evaluate reliability, interference, and power in LEO LR-FHSS uplink.
- Spatial scope: stepwise satellite movement over covered countries.
- Output metrics: decoded payload, collision rate, SNR/SINR ECDF, demod power.

$$\mathcal{Y} = \{Y_{c,t}, \gamma_t, \mathrm{SNR}_u, \mathrm{SINR}_u, P_{\mathrm{total}}(t)\}$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP.

End-to-End Flow

Orbit -> Coverage -> Load -> LR-FHSS -> Metrics

1. Orbit propagation
2. Coverage overlap
3. Population-to-load conversion
4. LR-FHSS decoding mapping
5. Collision/power/channel statistics

Flow With Stage Outputs

- Orbit output: satellite position and elevation per step
- Coverage output: country overlap fraction per step
- Load output: devices and demodulators per step
- LR-FHSS output: decoded payload by country-step
- Metric output: collision ECDF, SNR/SINR ECDF, power trace

System Model (Explicit)

- Single LEO satellite, multi-beam footprint, time-indexed by step (t)
- Countries/users inside footprint generate LR-FHSS uplink traffic
- Stepwise pipeline: geometry (\rightarrow) load (\rightarrow) decoding (\rightarrow) power/quality

$$\mathcal{S}(t) = \{\mathbf{r}_s(t),\ \mathcal{C}(t),\ N_{\mathrm{dev}}(t),\ D(t),\ Y(t)\}$$

Ref: "Fundamentals of Astrodynamics and Applications" - D. A. Vallado; "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021).

Channel Model (Explicit)

$$L_{u,\mathrm{tot}} = 20\log_{10}\left(\frac{4\pi d_u f_c}{c}\right) + A_{\mathrm{atm},u}$$

Ref: "A Note on a Simple Transmission Formula" - H. T. Friis (1946); "Propagation data and prediction methods for Earth-space links (P.618)" - ITU-R; "Attenuation by atmospheric gases (P.676)" - ITU-R.

$$\mathrm{SINR}_u = \frac{|h_{u,b_u^\star}|^2}{\sum_{b \neq b_u^\star} |h_{u,b}|^2 + N}$$

Ref: "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP.

Satellite-Earth Distance

- Model parameter: (h \approx 600\ \mathrm{km}) (LEO altitude)
- Earth radius: (R_E \approx 6371\ \mathrm{km})
- Near-zenith slant range: (d \approx h \approx 600\ \mathrm{km})

$$d(\psi) = \sqrt{(R_E + h)^2 + R_E^2 - 2R_E(R_E + h)\cos\psi}$$

Ref: "Fundamentals of Astrodynamics and Applications" - D. A. Vallado; "Orbital Mechanics for Engineering Students" - H. D. Curtis.

$$d_{\text{horizon}} \approx \sqrt{(R_E + h)^2 - R_E^2} \approx 2830\ \text{km} \quad (h = 600\ \text{km})$$

Ref: "Fundamentals of Astrodynamics and Applications" - D. A. Vallado.

Task 1: Orbit Model

$$a = R_E + h, \quad n = \sqrt{\mu/a^3}, \quad M(t) = M_0 + n(t - t_0)$$

Ref: "Astronomia Nova" - Johannes Kepler (1609); "Harmonices Mundi" - Johannes Kepler (1619); "Fundamentals of Astrodynamics and Applications" - D. A. Vallado; "Orbital Mechanics for Engineering Students" - H. D. Curtis.

$$M = E - e \sin E, \quad \mathbf{r}_{ECI} = R_3(\Omega) R_1(i) R_3(\omega) \mathbf{r}_{PQW}$$

Ref: "Astronomia Nova" - Johannes Kepler (1609); "Fundamentals of Astrodynamics and Applications" - D. A. Vallado; "Orbital Mechanics for Engineering Students" - H. D. Curtis.

Kepler Law Basis (Explicit)

$$T^2 = \frac{4\pi^2}{\mu} a^3, \qquad n = \frac{2\pi}{T}$$

Ref: "Harmonices Mundi" - Johannes Kepler (1619); "Philosophiae Naturalis Principia Mathematica" - Isaac Newton (1687); "Fundamentals of Astrodynamics and Applications" - D. A. Vallado.

$$r = a(1 - e\cos E), \qquad M = E - e\sin E$$

Ref: "Astronomia Nova" - Johannes Kepler (1609); "Orbital Mechanics for Engineering Students" - H. D. Curtis.

Task 1: Orbit Defensibility

- Inputs: orbital elements, Earth constants, time step.
- Processing: Kepler propagation + Earth rotation alignment.
- Outputs: track, elevation, orbit summary parameters.
- Why defendable: canonical aerospace equations, transparent assumptions.

$$\mathbf{r}_{ECEF}(t) = R_3(\omega_E t)\mathbf{r}_{ECI}(t)$$

Ref: "Fundamentals of Astrodynamics and Applications" - D. A. Vallado.

Task 2: Coverage Geometry

$$\xi_c(t) = \frac{A_{\mathrm{overlap},c}(t)}{A_c}, \quad 0 \le \xi_c(t) \le 1$$

Ref: "Virtues of the Haversine" - R. W. Sinnott (Sky and Telescope, 1984).

$$P_{\mathrm{eff}}(t) = \sum_c P_c \, \xi_c(t)$$

Ref: "Virtues of the Haversine" - R. W. Sinnott (distance-geometry basis for great-circle calculation).

Task 2: Load Construction

$$N_{\text{dev}}(t) = \rho\, P_{\text{eff}}(t), \quad D(t) = \left\lceil \frac{N_{\text{dev}}(t)}{C_{\text{dev/demod}}} \right\rceil$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021).

$$s_c(t) = \frac{P_c \xi_c(t)}{\sum_j P_j \xi_j(t)}$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021), system-level proportional resource allocation context.

Task 3: Link Budget Model

$$L_{u,\mathrm{FSPL}} = 20\log_{10}\left(\frac{4\pi d_u f_c}{c}\right)$$

Ref: "A Note on a Simple Transmission Formula" - H. T. Friis (Proceedings of the IRE, 1946).

$$L_{u,\mathrm{tot}} = L_{u,\mathrm{FSPL}} + A_{\mathrm{atm},u}$$

Ref: "Propagation data and prediction methods for Earth-space links (P.618)" - ITU-R; "Attenuation by atmospheric gases (P.676)" - ITU-R; "ITU-Rpy" - I. del Portillo et al.

Task 3: Noise and SNR

$$N = k_B T\, B\, F, \quad \mathrm{SNR}_u = \frac{P_{u,\mathrm{sig}}}{N}$$

Ref: "A Mathematical Theory of Communication" - C. E. Shannon (1948); "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP.

- Inputs: bandwidth, temperature, receiver noise figure.
- Output: user-level SNR baseline per step/elevation.

Task 4: Multi-Beam Channel and SINR

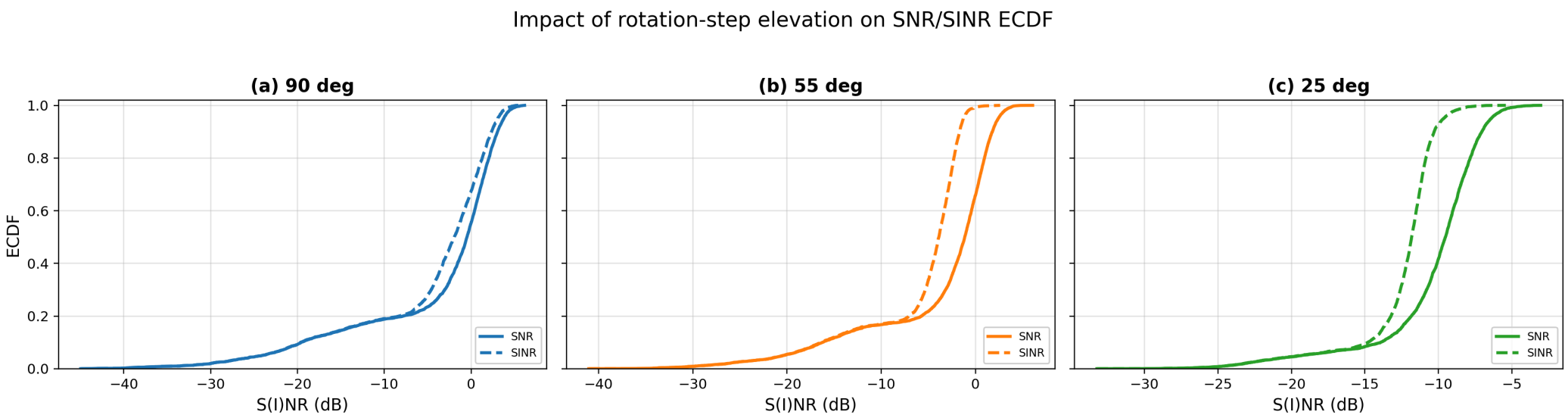
$$h_{u,b} = \sqrt{P_t \, 10^{(G_{\text{ant}} - L_{u,\text{tot}})/10}} \, g_u \, \mathbf{a}_u^H \mathbf{w}_b$$

Ref: "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP; "ITU-Rpy" - I. del Portillo et al.

$$b_u^\star = \arg\max_b |h_{u,b}|^2, \quad \text{SINR}_u = \frac{|h_{u,b_u^\star}|^2}{\sum_{b \neq b_u^\star} |h_{u,b}|^2 + N}$$

Ref: "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP.

Related Plot: Elevation vs SNR/SINR



- Three target elevations (25, 55, 90 deg) show clear distribution shift.
- Gap between SNR and SINR indicates interference impact.

Task 5: LR-FHSS Decoding Surrogate

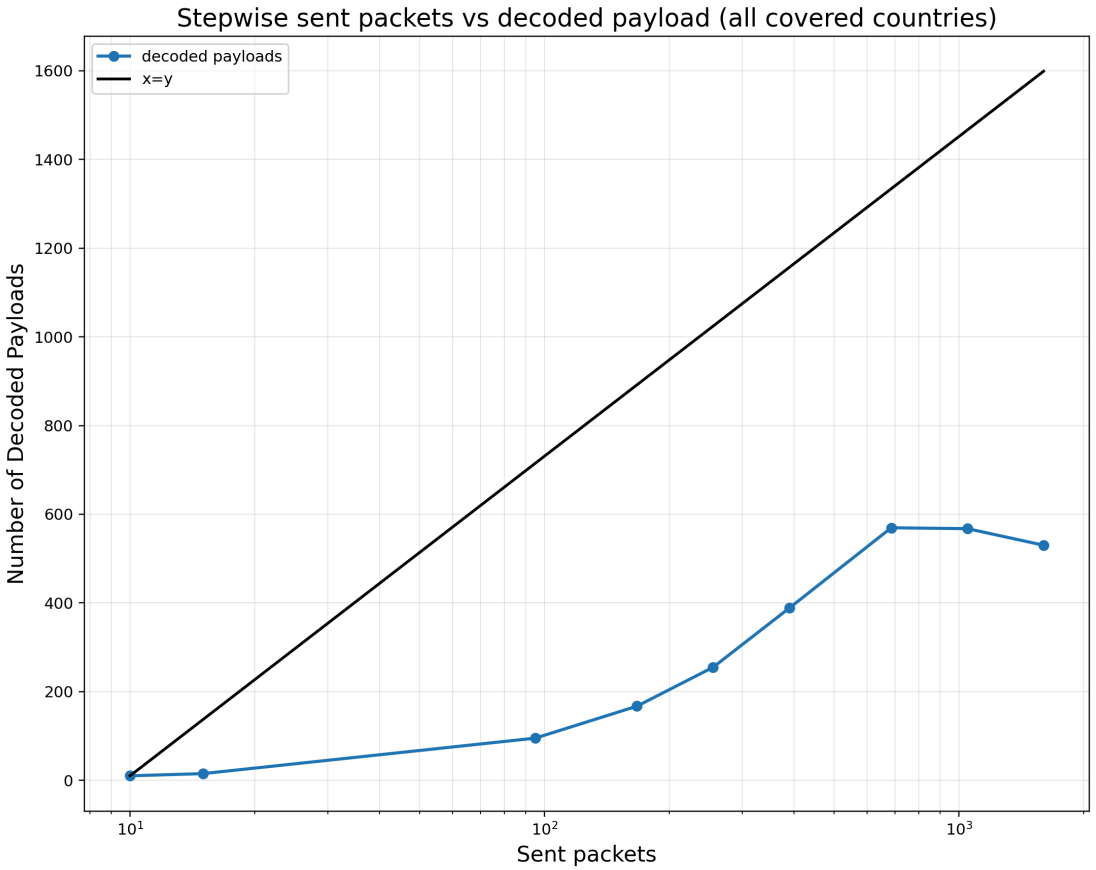
$$Y_{c,t} = f_{d_{c,t}}(n_{c,t})$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation" - Diego's paper (source of data-25dc-cr1.csv reference curves).

$$\eta_{c,t} = Y_{c,t}/n_{c,t}, \quad \gamma_{c,t} = 1 - \eta_{c,t}$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "The ALOHA System" - N. Abramson (1970).

Related Plot: Sent Packets vs Decoded Payload



- Points below $x = y$ indicate decoding loss under contention.
- Slope/dispersion reflect load-dependent LR-FHSS behavior.

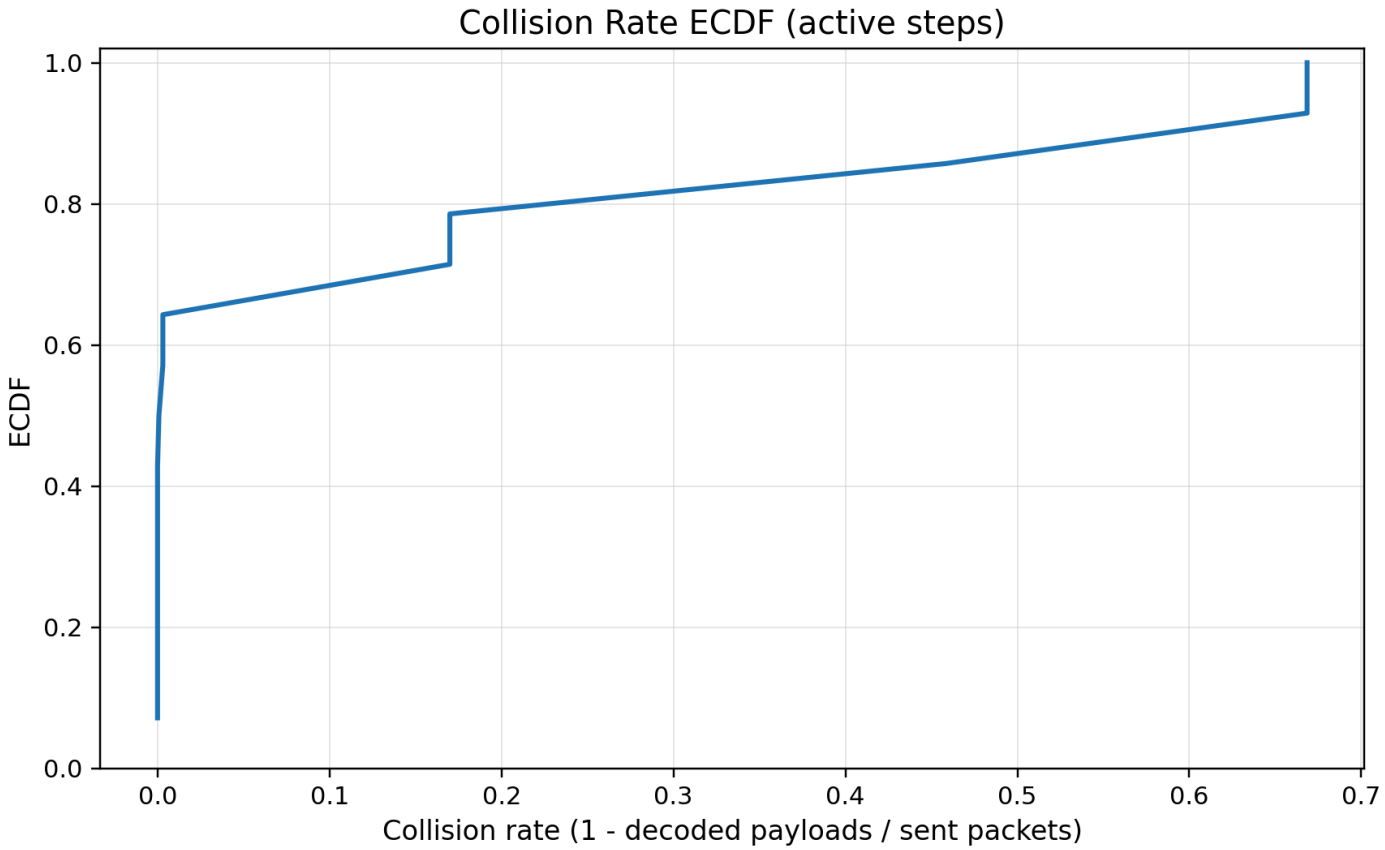
Task 5: Step-Aggregated Collision

$$\gamma_t = 1 - \frac{\sum_c Y_{c,t}}{\sum_c n_{c,t}}$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation" - Diego's paper (source of data-25dc-cr1.csv reference curves).

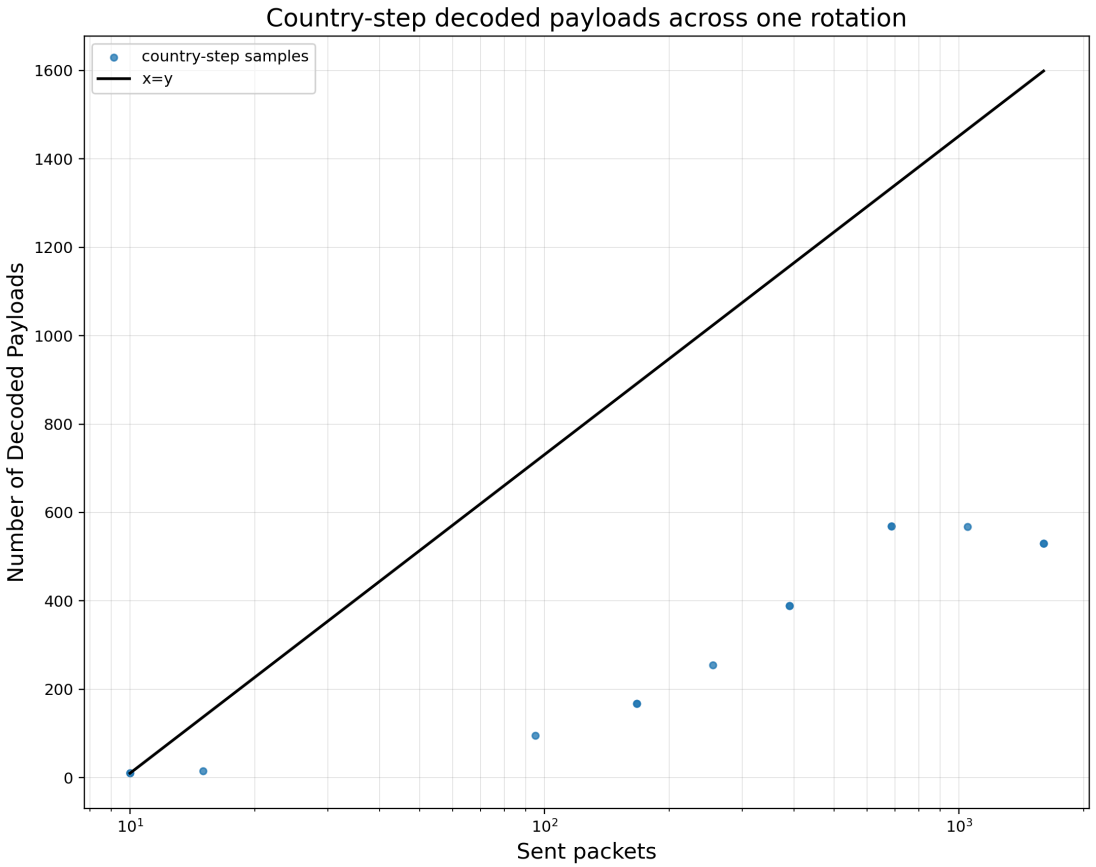
- Output type: per-step collision series and ECDF.
- Interpretation: captures congestion under time-varying coverage.

Related Plot: Collision ECDF



- Distribution view of collision risk across active steps.

Related Plot: Country-Level Sent vs Decoded (Average)



- Country-level average behavior against the (x=y) reference line.

Task 6: Demodulator Power

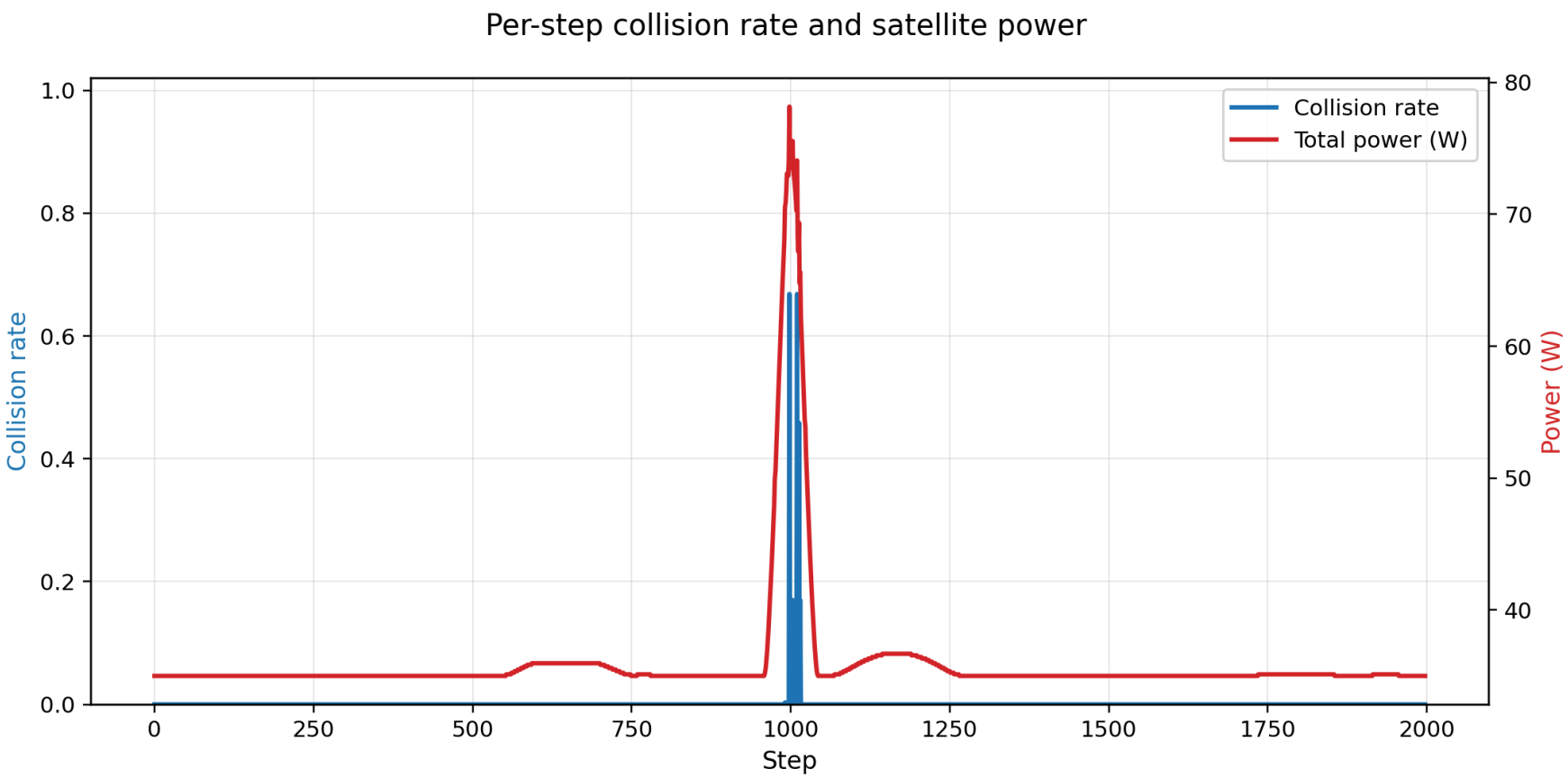
$$N_{\text{busy}}(t) = \min\left(D(t), \left\lceil \frac{\text{TX}(t)}{C_{\text{tx/demod}}} \right\rceil\right), \; N_{\text{idle}}(t) = D(t) - N_{\text{busy}}(t)$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation" - Diego's paper (source of data-25dc-cr1.csv reference curves).

$$P_{\text{total}}(t) = P_0 + N_{\text{idle}}(t)P_{\text{idle}} + N_{\text{busy}}(t)P_{\text{busy}}$$

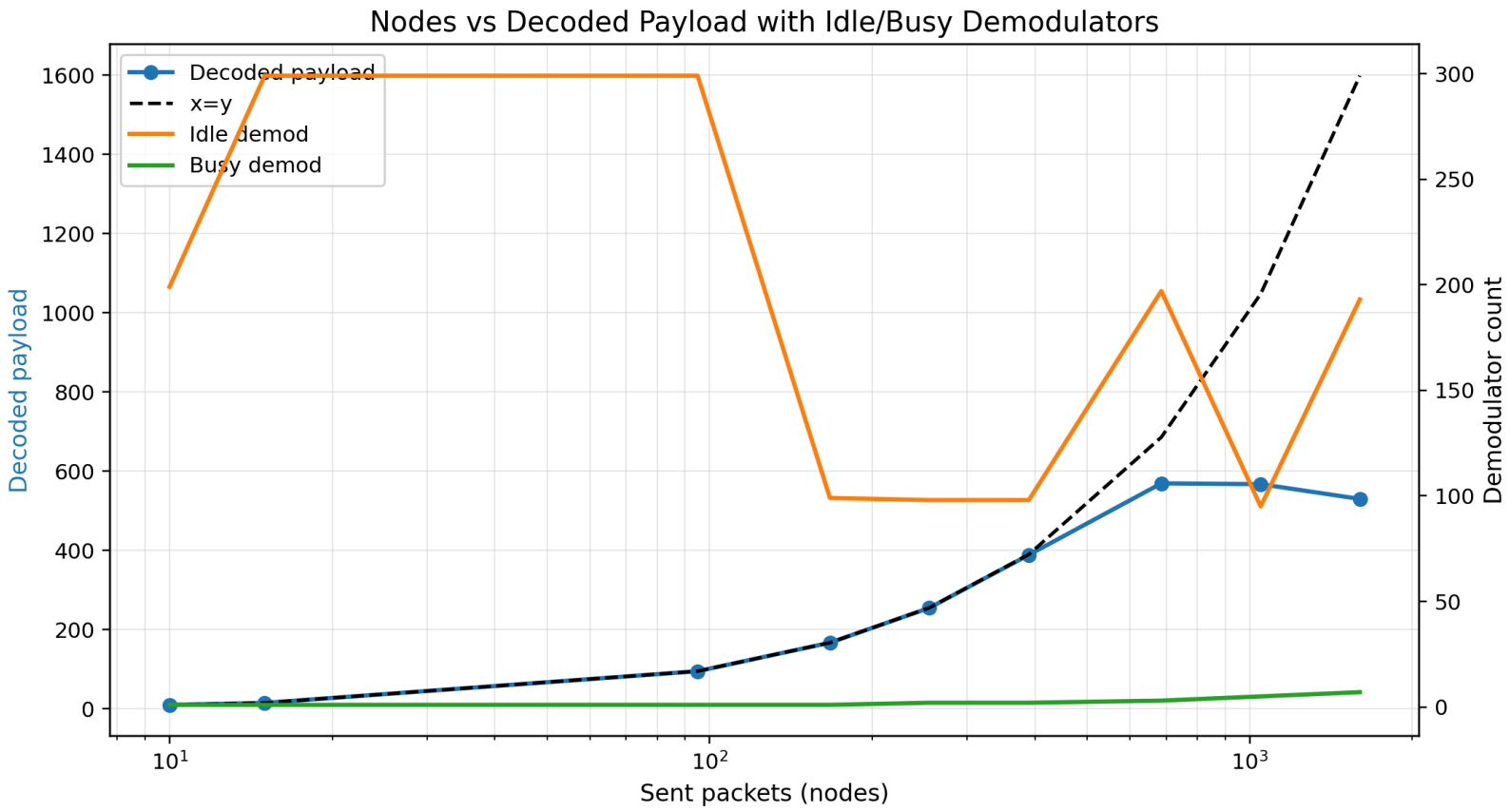
Ref: "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP.

Related Plot: Collision and Power Time Series



- Reliability-energy tradeoff over time.

Related Plot: Nodes vs Decoded Payload and Demod States



- Load scaling and demodulator-state evidence in one view.

Statistical Reporting

$$\hat{F}_X(x) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\{X_i \leq x\}$$

Ref: "All of Statistics" - L. Wasserman (Springer, 2004).

- Report ECDF for:
  - collision rate
  - SNR at 25/55/90 deg
  - SINR at 25/55/90 deg

Defensibility Boundaries

- Strength: all stages use explicit equations and cited models.
- Limitation: LR-FHSS stage uses surrogate curves (not full PHY per step).
- Limitation: country-level abstraction (centroid/area-equivalent overlap).
- Claim scope: comparative system behavior under stated assumptions.

Do not over-claim absolute field performance without calibration data

Ref: "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Solutions for NR to support NTN (TR 38.821)" - 3GPP; "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "Fundamentals of Astrodynamics and Applications" - D. A. Vallado.

Key Takeaways

1. The workflow is mathematically consistent from orbit to payload to power.
2. Each formula is grounded in standard references or explicit surrogate data.
3. The framework is suitable for defendable comparative analysis.

$$\text{Objective: maximize decoded payload while controlling collision and power}$$

Ref: "LR-FHSS: Overview and Performance Analysis" - N. Boquet et al. (2021); "Study on NR to support non-terrestrial networks (TR 38.811)" - 3GPP; "Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation" - Diego's paper (source of data-25dc-cr1.csv reference curves).

References

- "Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation" - (PDF manuscript provided by user).
- "Impact of Elevation Angle on Multi-Beam LEO Satellite Communication Systems" - (PDF manuscript provided by user).
- "LR-FHSS Overview and Performance Analysis" - (PDF manuscript provided by user).
- N. Boquet et al., "LR-FHSS: Overview and Performance Analysis," IEEE Communications Magazine, 2021.
- H. T. Friis, "A Note on a Simple Transmission Formula," Proc. IRE, 1946.
- C. E. Shannon, "A Mathematical Theory of Communication," BSTJ, 1948.
- J. Kepler, *Astronomia Nova*, 1609.
- J. Kepler, *Harmonices Mundi*, 1619.
- I. Newton, *Philosophiae Naturalis Principia Mathematica*, 1687.
- ITU-R P.618, Earth-space propagation prediction methods.
- ITU-R P.676, Attenuation by atmospheric gases.
- 3GPP TR 38.811, Study on NR to support non-terrestrial networks.

References (Cont.)

- D. A. Vallado, *Fundamentals of Astrodynamics and Applications*, 4th ed.
- H. D. Curtis, *Orbital Mechanics for Engineering Students*, 4th ed.
- R. W. Sinnott, "Virtues of the Haversine," *Sky and Telescope*, 1984.
- I. del Portillo et al., *ITU-Rpy* (implementation of ITU-R attenuation models).
- N. Abramson, "The ALOHA System," AFIPS, 1970.
- L. Wasserman, *All of Statistics*, Springer, 2004.
- LR-FHSS reference dataset: `data-25dc-cr1.csv` (driver/lifan curves), sourced from "Enhancing LR-FHSS Scalability Through Advanced Sequence Design and Demodulator Allocation" (Diego's paper).